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Didactic transposition and the knowledge to be taught: towards an archeorganisation for concave/convex functions

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ABSTRACT

This study is framed within the anthropological theory of the didactic. The first part is a comparative study of the mathematical organisation of concave/convex functions on the one hand at university, and on the other hand in secondary education. Data for this analysis are from a university textbook on calculus and a secondary school textbook. The analytic results show how the mathematics (to be) taught at secondary level consists of a largely demathematised *logos* block. The corresponding institutional fragmentation of knowledge reveals a missing praxeological organisation from which the organisations of knowledge we consider in this paper could be derived by transposition. Responding to this insight, in the second part of the paper, we have outlined what we call an *archeorganisation* for concave/convex functions, a framework that teacher education institutions could refer to when trying to satisfy the specific constraints they are subjected to (notably because of its publics), without for that matter isolating themselves epistemically from other academic institutions. Formulating an archeorganisation represents a meta-curricular study, with the objective of transforming the institutional conditions and constraints that guide mathematics educators and curriculum designers in their deliberations about the concept of concave/convex functions.

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Archeorganisation; calculus; concavity and convexity; fragmentation and defragmentation of knowledge; institutional and didactic transposition; mathematical organisation; mathematics teacher education; mathematics textbook; praxeology; the anthropological theory of the didactic

1. Introduction

Anchored within the framework of the Anthropological Theory of the Didactic (ATD), this study begins with a comparative analysis of the exposition of concave and convex functions within two mathematics textbooks, one tailored for university and the other for secondary education. The objective of this analysis is to describe and explain the institutional conditions and constraints imposed on mathematics students and educators concerning concave and convex functions, with a view to making it known to the broader scholarly community. The query that initiated the study was raised by a student teacher, directed to one of the authors of this paper. It concerned the *logos* block (i.e. roughly speaking, the mathematical

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theory) relating to the concavity of functions in a Grade 12 textbook, and was connected to the student teacher's master's thesis study (Kristianslund, 2022).

The comparative analysis shows how the *logos* block of concave/convex functions in the secondary education textbook bears indications of transformations due to didactic transposition processes. This led us to develop a reference model – a praxeological organisation which we call an *archeorganisation* – as a tool to guide the analysis and design of didactic transposition processes pertaining to the notion of concavity, particularly in teacher education institutions. The creation of an archeorganisation can be considered a meta-curricular study. What we try to change for readers – primarily educators and curriculum designers in mathematics – is the set of institutional conditions and constraints that inform curricular deliberations on the subject of concave and convex functions.

The existing body of didactic research contains a relatively limited quantity of studies focusing on the concavity of functions. While it is not within our scope to provide an exhaustive review of the entirety of this corpus, we intend to present a selective synopsis of this body of literature in the subsequent paragraphs.

Published research in mathematics education concerning the teaching and learning of concavity/convexity and inflection points has shown a notable emphasis on students' 'mathematical thinking' about concavity, that is, in ATD terms, on students' relation to concavity. This body of work frequently uses widespread, cognitive theoretical frameworks, with significant contributions from researchers such as Fischbein, Tall, Vinner and Duval, who have explored the nature of 'mathematical understanding' and concept formation. For instance, Carlson et al. (2002) found that calculus students struggle with understanding and depicting concavity and inflection points on graphs. When asked to explain these concepts in a dynamic scenario, students often resorted to rather simplistic statements like 'when the second derivative is positive, the graph is concave up' or 'when the second derivative is equal to zero, there is an inflection point' (Carlson et al., 2002, p. 355, our paraphrasing). Further inquiries revealed that these students seemed to lack comprehension of the principles underlying the 'methods' they used. In ATD terms, these students demonstrate a limited *logos* component of their praxeology relating to concavity and inflection points. This deficiency hinders their capacity to adequately explain and justify or refute the techniques they employ. Overall, Carlson et al. (2002) found that participants in their study did not demonstrate behaviours indicative of understanding 'covariational reasoning' (i.e. how two variables change in relation to one another).

Tsamir and Ovodenko (2013) similarly explored the challenges university students face in relating to the concept of inflection point. One key finding is that students often mistakenly believe that the condition $f'(x) = 0$ is necessary and that $f''(x) = 0$ is sufficient for identifying inflection points. The authors advocate for the use of diagnostic tools such as questionnaires to detect student misconceptions early. Encouraging students to engage in reflective thinking and to adhere to exact definitions when performing mathematical tasks is emphasised as a key strategy.

A study by Jones (2019) points at the significance of covariational reasoning abilities in their relation to real-world quantities. He suggests that explicitly discussing the covariation of real-world quantities with calculus students in relation to concavity and inflection points, while adhering to the mathematical definitions of first and second derivatives, may better equip them to handle these crucial concepts. While Carlson et al. (2002) primarily focus on deficiencies in students' abilities to perform tasks on concavity and inflections

points, Jones (2019) advances the discussion by advocating the use of these concepts in real-life scenarios, ensuring that they are anchored in the ‘underlying’ mathematical concepts. This emphasis on mathematical definitions is not unsimilar to the stance adopted by Tsamir and Ovodenko (2013) for the concept of inflection point. From the perspective of the ATD, Jones’s (2019) advocacy for altered didactic conditions (and constraints) for engaging with these concepts is apparent. In terms of research, his proposal is anticipated to stay theoretically within the cognitive domain, with data presumably to be derived from participant interactions in the mathematics classroom.

Our research study is about the institutional conditions and constraints under which actors study and teach (i.e. contribute to study) the notion of concavity of functions. From this point of view, we endeavour to make a contribution to the exploration of the following research questions:

RQ1. ‘In which ways does the version of concave/convex functions in a mathematics textbook for secondary education differ from the version in a calculus textbook for university?’

RQ2. ‘How can the discrepancies due to the institutional fragmentation of knowledge caused by the processes of didactic transposition be reduced by constructing an archeorganisation recognised as such by the institutions concerned with mathematics education, in particular teacher education institutions?’

The second research question emerged from the insights gained through the investigation of the first. We examine RQ1 and RQ2 in Sections 3 and 4, respectively. Prior to delving into these analyses, we present the theoretical framework of the study in the next section. The paper culminates with a discussion and conclusive remarks in Section 5.

2. Theoretical framework: some key concepts from the ATD

The object of this section is to expound upon the foundational concepts that our study draws upon, including praxeology, institutional and didactic transposition and arch-eschool.

2.1. A conceptual update

Yves Chevallard’s book *La transposition didactique. Du savoir savant au savoir enseigné* (‘Didactic transposition: From scholarly knowledge to knowledge taught’), first published in 1985, was based on a series of lectures given in 1980 in France as part of the first Summer School on the Didactics of Mathematics. A second, expanded, edition, was published in 1991 (Chevallard, 1991). A key concept was the notion of a ‘body of knowledge’. The mathematical theory of rational numbers, of quadratic equations, of systems of linear equations, of conic sections, of geometric transformations, of probability calculus are some examples of such bodies of (mathematical) knowledge. That was some forty years ago. Since then, and with the theory of didactic transposition as a starting point, the ATD (Chevallard, 2019; *in press*) has gradually evolved to reach its current state of development. In the course of this process, the notion of a body of knowledge was defined more precisely with the help of the notion of *praxeology* (Chevallard, 2020; Chevallard & Bosch, 2020a). The theory of praxeologies postulates that any relatively stabilised human activity comes down to carrying out, in a more or less predetermined sequence, a set of *tasks* t_i of types T_i . Moreover,

performing a task t of type T amounts to implementing a certain *technique* τ . The ordered pair formed by T and τ , traditionally denoted by $[T/\tau]$, is called the *praxis* block. Generally speaking, a given technique can competently perform only *some* of the tasks t_i of types T_i : these make up the *scope* of the technique.

Another anthropological postulate of the theory of praxeologies is as follows: each of these ways of doing things, which are the techniques τ , must be explained, justified and made intelligible in the institutions where these techniques are used. This is the role played by the so-called *technology* θ of τ . The word ‘technology’ is to be understood here as referring to a *discourse* (*logos*), regarded as rational in the institutions in which it is uttered, whose object is the technique τ (*tekhnē*). As a general rule, θ is not fully self-justified: some other principles must be added to it. These principles are often left more or less implicit in the institutions concerned; they are of a higher level of generality and makeup what is known as the *theory* Θ (in ancient Greek, *theōrein* meant ‘to consider, speculate, look at’, and *theōros* meant ‘spectator’). The pair $[\theta/\Theta]$ formed by θ and Θ is called the *logos* block. The technology θ and, more broadly, the *logos* block $[\theta/\Theta]$ do not only have the role of justifying the technique: in a number of cases, particularly in mathematics, the *logos* block is supposed to *generate* the technique, at least in the eyes of the institution where all this takes place. It is essential to acknowledge that the *role* of the *logos* block in one institution may be different from its role in another institution: this discrepancy is one of the problems raised by didactic transposition processes. That noted, a praxeology ρ is the union of a *praxis* block and a *logos* block and is traditionally written as follows: $\rho = [T/\tau/\theta/\Theta]$. As we already did, it is conventional to designate a praxeology by referring to its *theoretical* component Θ , thus speaking of the *theory of...* (catastrophes, categories, graphs, numbers, rings, equations, etc.). Our use of the term ‘theory’ must then be understood to be synecdochic, that is, to designate a totality (the praxeology in question) by one of its parts (its theory) deemed most representative. Finally, it cannot be emphasised too strongly that, just as a technique may have a very limited scope, similarly a technology (and more broadly a *logos* block) can have limited explanatory, justificatory or generative power.

So what is a body of knowledge? It is a praxeology or, more often, a more or less integrated *system* (or *organisation*) of praxeologies. The partial integration of two praxeologies $\rho = [T/\tau/\theta/\Theta]$ and $\rho' = [T'/\tau'/\theta'/\Theta']$ can be achieved in particular at the level of *logos* (θ and θ' can share certain elements), and even more at the theoretical level, where Θ and Θ' can share even more decisive elements. In what follows, we will call ‘element of knowledge’ or ‘praxeological element’ any part of a body of knowledge, that is to say any part of a praxeological system. We will now move on to the notion of transposition.

2.2. Institutional and didactic transposition processes

Formulated within the current framework of the ATD, the theory of didactic transposition arose from the following question: Given an institution I and a body of knowledge $\mathcal{K}$ or, more precisely, an element of knowledge $\mathcal{K}$ implemented in I , where does $\mathcal{K}$ come from? Observation shows that, as a general rule, while very few components of $\mathcal{K}$ have been created in the institution I itself, most of it generally comes from a number of institutions which, at some point in their history (centuries ago, decades ago, some years ago, etc.), carried within them some of the constituents from which $\mathcal{K}$ was derived in I . The process by which an element of knowledge $\mathcal{K}$ moves from one institution to another, not without

changes, is called a process of *institutional transposition* (Chevallard & Bosch, 2020b). The reason for the changes that occur is that the conditions and constraints under which $\mathcal{K}$ lived in the ‘home’ institutions differ significantly from the conditions and constraints under which $\mathcal{K}$ should live in the institution of arrival, I . Just think for example of the difference between the mathematical elements produced by mathematicians such as Descartes, Abel, Galois, Poincaré or Hilbert, among so many others, which constitute *scholarly knowledge* in the strict sense of the term, and the knowledge taught even in the first years of university. In the first case, the mathematical knowledge produced is didactically almost untreated, which is obviously *not* the case in the second instance – hence the name *didactic transposition* given to the institutional transposition when the institution I regards $\mathcal{K}$ as ‘knowledge to be taught’.

Given a pair $(I, \mathcal{K})$, where $\mathcal{K}$ is a praxeological element living in the institution I , we can look for institutions $\bar{I}$ containing a praxeological element $\bar{\mathcal{K}}$ which is an *ancestor* of $\mathcal{K}$ in I , that is, an element of knowledge of which certain constituents contributed to the making of $\mathcal{K}$ in I , $\mathcal{K}$ being called a *descendant* of $\bar{\mathcal{K}}$. The n -tuple $(\bar{I}_1, \bar{I}_2, \dots, \bar{I}_n)$ of ancestors of $\mathcal{K}$ in I arranged according to historical chronology is called the *genealogy* of $\mathcal{K}$. Given two pairs $(I, \mathcal{K})$ and $(I', \mathcal{K}')$, it is often very enlightening to carry out a *comparative praxeological analysis* of $\mathcal{K}$ and $\mathcal{K}'$, especially when it is not unreasonable to think that $(I', \mathcal{K}')$ might be one of the ancestors of $(I, \mathcal{K})$. This is what we shall do in Section 3, where we investigate our first research question, RQ1, introduced in Section 1.

2.3. The notion of archeschool

Let us consider an institution I and its *praxeological needs*, that is, its need for tools for action and thinking about action. Naturally, such needs depend on the activities developed or to be developed within I . Suppose the actors in control of I judge that an existing or future activity in I necessitates the fulfilment of new praxeological needs. What can the institution I do under these conditions? Do not forget that what is called an institution here is just as much a secondary school *class* as it is a university *department* or a commercial *company*, and so on. On a permanent or temporary basis, a sub-institution is formed in I , which we call the *noosphere* of I , $\mathcal{N}_I$, that is, the ‘place’ where people ‘think’ about the dynamics of I and the decisions it calls for (see, e.g. Merriam-Webster, n.d.c). In the simplest case, $\mathcal{N}_I$ will be looking for an institution $\bar{I}$ that satisfies the following two conditions: firstly, $\bar{I}$ seems to carry within it the desired praxeological elements; secondly, $\bar{I}$ enjoys sufficient *epistemic legitimacy* for I to be able to repel any attack (erratic or systematic) on an element $\mathcal{K}$ transposed from $\bar{I}$ that would call into question the epistemic validity of $\mathcal{K}$.

There is often a compromise to be found between the praxeological utility for I of what $\bar{I}$ can offer and its epistemic legitimacy. A well-known example is the ‘new math’ reform, but we will not go any further into that here (see Chevallard, 1991, pp. 24–28). The noosphere $\mathcal{N}_I$ designates as ‘knowledge to be taught’ the elements of knowledge that it has identified in $\bar{I}$ as being relevant for I . Another sub-institution of I is needed to ensure the transposition of $\mathcal{K}$ into I : a *school* $\mathcal{S}_I$, where the subjects of I who need to master $\mathcal{K}$ will come to *study* $\mathcal{K}$. The word ‘school’ comes from the Greek *skholē*, which originally meant ‘spare time, leisure’, and which later came to mean ‘time devoted to study taken from the institution’s time’. A school, in the sense in which this word is used in the ATD, is thus any place and time devoted to study. Let us consider a limiting case. An 11th-grade teacher $\hat{y}$ has to teach

a new part of the curriculum, which he has never taught or even studied before – let us say, rudiments of graph theory, for example. We also suppose that, in this institution I that is $\hat{y}$'s classroom, $\hat{y}$ is the unique member of the noosphere $\mathcal{N}_I$ ($\hat{y}$ is 'independent' and does not like to collaborate with colleagues).

Moreover, $\hat{y}$ creates his own school $\mathcal{S}_I$ by using part of his spare time to study a book on graph theory he has come across. Let us now imagine a set of 11th-grade classes, whose teachers have to tackle the same problem under the same conditions. Instead of each teacher doing his or her own thing, they can have recourse to a joint school $\hat{S}$, the purpose of which is to provide them with the necessary education. $\hat{S}$ can be identified, concretely, with teacher education institutions. In a given country such as Norway or France, there is a number of such institutions $\hat{S}$ distributed over the national territory. In principle, each of them can provide their own answers to the questions posed by the teachers concerned – such as what are the praxeologies relating to graph theory that an 11th-grade teacher must master and teach –, while, while satisfying the conditions of both *praxeological utility* and *epistemic legitimacy* that we have discussed.

What can happen, however, is that these institutions $\hat{S}$, behaving rather like the 'individualistic' professor mentioned above, do not agree on the answers they come up with and do not discuss them among themselves, in particular for lack of an institutional place and time to do so. Our study is placed in the perspective of the construction of such a forum, the existence of which should make it possible to increase the qualities of praxeological utility and epistemic legitimacy of the answers provided in the various teacher education institutions $\hat{S}$. We call such an institution an *archeschool* and denote it by $\check{S}$. The notion of archeschool was introduced in Michèle Artaud's doctoral dissertation (Artaud, 1993; see also Artaud, 2021; and Artaud & Bourgade, 2022). It should be remembered here that the prefix *arche-* (as for example in 'archetype') comes from the Greek *arkhē* 'beginning, origin, first place' (for more on this, see Ayto, 2005, p. 33.). An 'archeschool' is thus a 'primordial' school, that is, a school that serves as a reference for particular schools which recognise it as their main source from the dual point of praxeological utility and epistemic legitimacy.

3. A comparative study of institutional outcomes of didactic transposition

The objective of this section is to explore our first research question, RQ1, introduced in Section 1. The analytic technique employed is essentially that of *didactic transposition analysis* (Chevallard, 1991). The didactic transposition analysis of the concerned body of knowledge – concave/convex functions, $\mathcal{K}$ – involves a comparison of praxeological analyses of two different 'copies' (i.e. 'transposed' versions) of $\mathcal{K}$ as they may appear in two different institutions: at the university and in secondary school. The empirical basis for the comparative analysis consists of two textbooks; these are introduced in the following subsection.

3.1. Sources of data

Our didactic transposition analysis is based on the examination of two textbooks.¹ The first, called *Calculus* (in Norwegian, *Kalkulus*), is an introductory textbook on calculus for the university, published in 2016. It is written by Tom Lindstrøm, professor of mathematics at the University of Oslo. The second is entitled *Patterns* (in Norwegian, *Mønster*), a

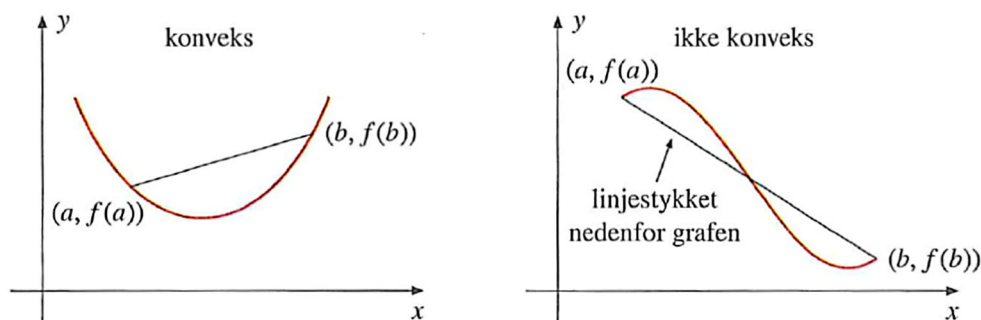


Figure 1. Convexity (Lindstrøm, 2016, p. 314). Reprinted with permission.

Grade 12 textbook for a ‘theoretical programme’ preparing for university studies in science, technology, engineering and mathematics. It is part of a textbook series for the national curriculum since 2020, written by Tove Kalvø, Jens C. L. Opdahl, Knut Skrindo and Øystein J. Weider, all serving as teachers of mathematics at different upper secondary schools. *Calculus* was chosen because it is an introductory textbook used at several Norwegian universities in the first calculus course taken by students enrolled in teacher education programmes for Grade 8–13. *Patterns* was selected on the account of being part of a brand-new textbook series for the theoretical programme; it is not a revised version of an old series as are, for example, Borgan et al. (2021) and Oldervoll et al. (2021) for the same programme.

In the subsequent subsections, we first give an account of how concave/convex functions is treated in the university textbook chosen, *Calculus*. We shall focus on its *logos* block, which we regard as closer to scholarly knowledge. Then we analyse how the same theme is outlined in the school textbook chosen, *Patterns*, and compare its outcome with the treatment in the university textbook. The aim is to bring to light the differences in the treatment of $\mathcal{K}$ in *Patterns*.

3.2. The logos block relating to concavity in the university textbook

In Chapter 6.4 of *Calculus*, under the heading ‘Discussion of Curves’, there is a section entitled ‘Convex and Concave Functions’ (pp. 313–318). The author starts with a geometrical definition of the concepts of *convex function* and *concave function*:

6.4.5 Definition. The function f is called *convex* on the interval I if every time we select two points $a, b \in I$, then no point on the line segment between $(a, f(a))$ and $(b, f(b))$ will be below the graph of $y = f(x)$ (see [Figure 1]).² We say that f is *concave* on I if every time we select two points $a, b \in I$, then no points on the line segment between $(a, f(a))$ and $(b, f(b))$ will be above the function graph (see [Figure 2]). (Lindstrøm, 2016, p. 314)

Lindstrøm (2016) continues to build up elements in the *logos* block, which allows him to deduce a connection between concavity and the second derivative of a function twice differentiable. To be able to use the ‘mean value theorem’³ in the proof of the theorem that establishes the sought relationship, the following lemma using difference quotients is presented (p. 315):

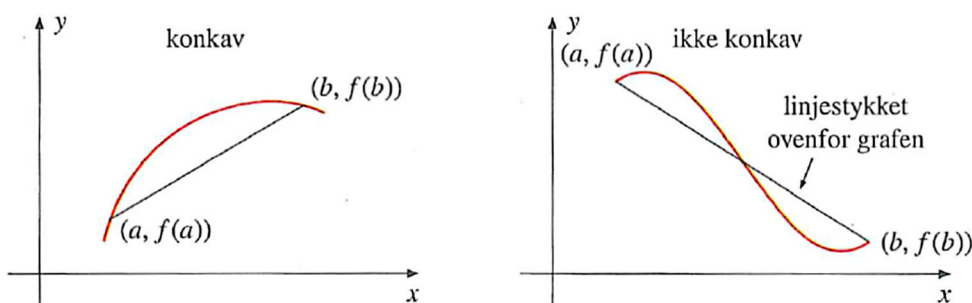


Figure 2. Concavity (Lindstrøm, 2016, p. 314). Reprinted with permission.

6.4.6 Lemma. A function is convex on an interval I if and only if the following applies. For all points $a, b, c \in I$ such that $a < c < b$, we have

$$\frac{f(c) - f(a)}{c - a} \leq \frac{f(b) - f(c)}{b - c} \quad (1)$$

Correspondingly, f is concave on I if and only if

$$\frac{f(c) - f(a)}{c - a} \geq \frac{f(b) - f(c)}{b - c}. \quad (2)$$

The author writes that the two statements can be proved the same way and presents a proof for the convexity part of the lemma:

Proof: Assume first that f is convex. Then the point $(c, f(c))$ cannot be above the segment connecting $(a, f(a))$ and $(b, f(b))$, and we must have the situation shown in [Figure 3]. When we compare the slopes k , k_1 and k_2 of the three segments in the figure, we see that $k_1 \leq k \leq k_2$, which means that

$$\frac{f(c) - f(a)}{c - a} \leq \frac{f(b) - f(a)}{b - a} \leq \frac{f(b) - f(c)}{b - c}$$

If we omit the middle part, we get (1). (Lindstrøm, 2016, p. 315)

The next step of the author consists in assuming that f is *not* convex and showing, consequently, that (1) does not hold:

Since f is not convex, we can find points $a, b, c \in I$ so that $a < c < b$ and $(c, f(c))$ is above the segment connecting $(a, f(a))$ and $(b, f(b))$; this means that we have the situation shown in Figure 4.

We now see that the ratio between the slopes is $k_1 > k > k_2$ – in other words

$$\frac{f(c) - f(a)}{c - a} > \frac{f(b) - f(a)}{b - a} > \frac{f(b) - f(c)}{b - c}$$

If we omit the middle part, we get the inverse inequality of (1). ■ (Lindstrøm, 2016, pp. 315–316)

Then everything is ready to state the theorem that is the central element in the author's mathematical organisation of the *logos* part on the concavity of functions:

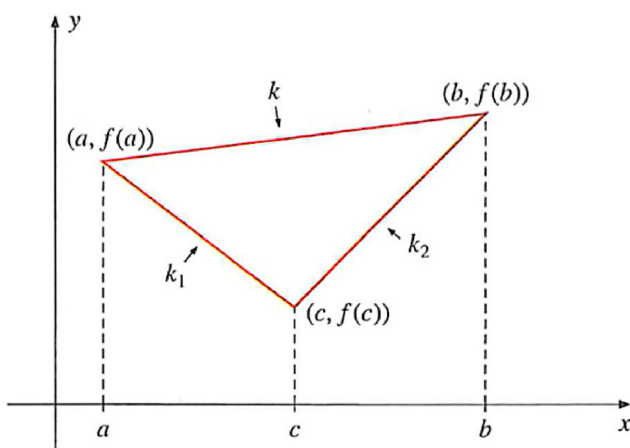


Figure 3. A convex function (Lindstrøm, 2016, p. 315). Reprinted with permission.

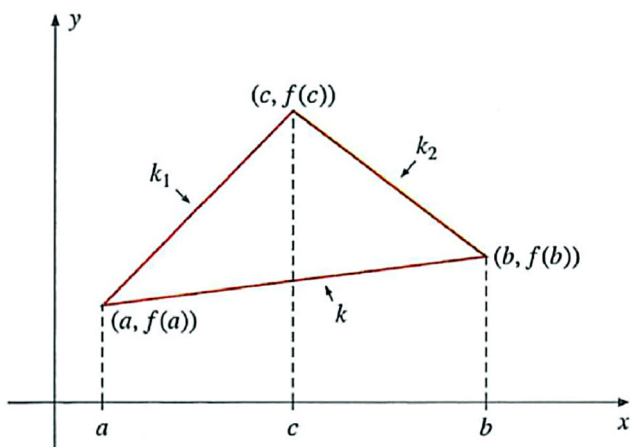


Figure 4. A concave function (Lindstrøm, 2016, p. 316). Reprinted with permission.

6.4.7 Theorem. Assume that f is continuous on an interval I and that $f''(x) \geq 0$ for all inner points $x \in I$. Then f is convex on I . If instead $f''(x) \leq 0$ for all inner points of I , then f is concave on I . (Lindstrøm, 2016, p. 316)

The proof addresses the convexity part of the theorem, using the above lemma:

Proof: Choose three points $a, b, c \in I$ so that $a < c < b$. According to Lemma 6.4.6, it suffices to prove that $\frac{f(c)-f(a)}{c-a} \leq \frac{f(b)-f(c)}{b-c}$. By the mean value theorem, there exist two numbers $c_1 \in (a, c)$ and $c_2 \in (c, b)$ so that $\frac{f(c)-f(a)}{c-a} = f'(c_1)$ and $\frac{f(b)-f(c)}{b-c} = f'(c_2)$. Since $f''(x) \geq 0$, f' is increasing and, consequently, $f'(c_2) \geq f'(c_1)$ [because $c_1 < c < c_2$]. Hence $\frac{f(c)-f(a)}{c-a} = f'(c_1) \leq f'(c_2) = \frac{f(b)-f(c)}{b-c}$. ■ (Lindstrøm, 2016, p. 316)

After this, two examples are given that discuss concavity. The second example introduces the notion of *inflection point* with this formulation: ‘ a is an inflection point for f if f is continuous at a and there exists an $\varepsilon > 0$ so that f is convex on one of the intervals $(a - \varepsilon,$

a), $(a, a + \varepsilon)$ and concave on the other' (Lindstrøm, 2016, p. 318). An inflection point is a point where a function changes from being concave to being convex or vice versa. This is succeeded by 18 tasks that address the shapes of curves more broadly.

The above is a brief account of the 'scientific' treatment of the concept of concavity in *Calculus*, which describes the constituent parts of the *logos* block relating to concavity: (1) a *definition* of a convex/ concave function; (2) the *mean value theorem* (proved in a previous section of the textbook) used in the proof of a lemma to be used in the proof of a central theorem with respect to the concept at stake; (3) the above-mentioned *lemma* (with proof); (4) the central *theorem* (with proof) declaring a connection between the sign of the second derivative and the concavity of a function; and (5) a *definition* of inflection point.

3.3. The notion of concavity of functions in the school textbook

Let us now consider the treatment of concave/convex functions in the Grade 12 textbook *Patterns* (Kalvø et al., 2021, pp. 208–224). First, our examination centres on the authors' communication style; second, we conduct a praxeological analysis of the theme, comparing its outline in the school textbook with its presentation in the university textbook, accounted for in the previous subsection.

3.3.1. Utilising colloquial expressions and graphical illustrations

A first remarkable point is that the section in *Patterns* devoted to concavity issues is entitled 'The Second Derivative'. The question of concave/convex functions is thus presented as an application of the notion of second derivative. A second remarkable point is that the words 'concave' and 'concavity' do not appear in *Patterns*: 'concave' is replaced by the expression 'hollow side' (in Norwegian, *hul side*), that is the side that faces either down or up; and 'concavity' is replaced by the term 'curvature' (in Norwegian, *krumning*). We will see that this is a 'symptom' of the treatment of concavity in this textbook. The mentioned notions appear in the following passage:

We compare this with the graph of f and see the following:

- When f'' is negative, f' is decreasing and the graph of f turns its *hollow side* down.
- When f'' is positive, f' is increasing and the graph of f turns its *hollow side* up.

A function with a graph turning its hollow side up or down is not linear. We say that the graph curves, and we mark the *curvature* of the graph with an arc below the sign line [see Figure 5]. (Kalvø et al., 2021, pp. 211–212)

Thus, words that are traditional in mathematics (concave/convex and concavity/convexity) but which, a priori, mean little or nothing to the students, are replaced by expressions that make sense in everyday language (i.e. hollow side facing down/up and curvature). These informal expressions have a metaphorical value and are implicitly used in the school textbook as *definitions*. The same is true for the notion of *inflection point* (in Norwegian, *infleksjonspunkt*) which the authors have replaced with the expression 'turning point' (in Norwegian, *vendepunkt*):

The point where the graph goes from facing the hollow side up to [facing] the hollow side down (or vice versa), is called the *turning point*. At the turning point, the sign of the second derivative changes. (Kalvø et al., 2021, p. 211)

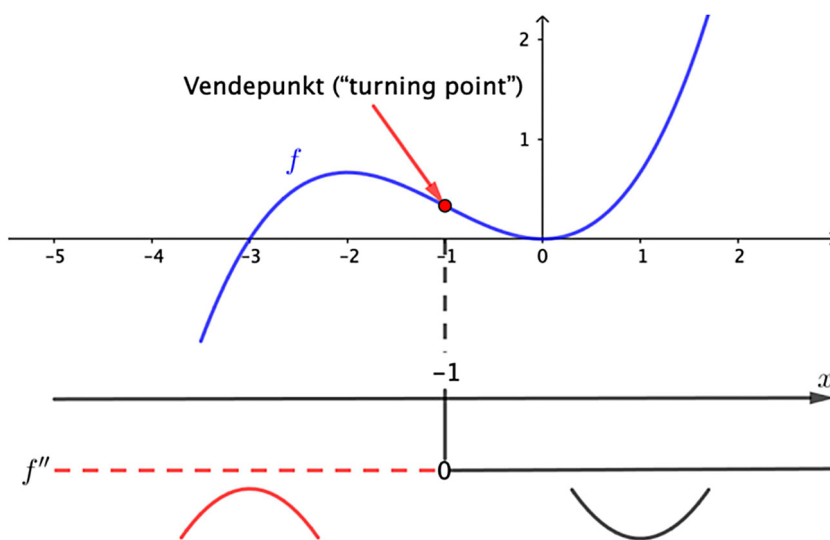


Figure 5. Sign line for the second derivative (adapted from Kalvø et al., 2021, p. 211). Reprinted with permission.

The term ‘turning point’ is widely recognised within the field of calculus as a point where the graph of a function changes direction from increasing to decreasing (signifying a local maximum) or from decreasing to increasing (signifying a local minimum). Hence, using ‘turning point’ to describe an inflection point could lead to confusion, given that an inflection point signifies a change in concavity rather than a local extremum.⁴ For instance, a definition given by Mercer (1910/1914, pp. 106–107) can be distilled as follows: *a* is a *turning point* for *f* if *f* is continuous at *a* and there exists an $\varepsilon > 0$ so that $f'(x) < 0$ on one of the intervals $(a - \varepsilon, a)$, $(a, a + \varepsilon)$ and $f'(x) > 0$ on the other (see also Perry & Perry, 1982, p. 242, and ‘Intercepts and Turning Points of Polynomial Functions,’ n.d.).

The *technology* of the technique of using a sign line for the second derivative is in fact reduced to a minimum. The authors of the textbook have adopted a ‘naturalistic’ approach to functions. They do so by considering a specimen function regarded as *generic*, in this case the function defined by $f(x) = \frac{1}{6}x^3 + \frac{1}{2}x^2$ for all $x \in \mathbb{R}$. The graphs of *f*, $f'(x) = \frac{1}{2}x^2 + x$ and $f''(x) = x + 1$ are shown in Figure 6.

Looking at the graph of *f*, we ‘see’ that *f* first increases, reaches a maximum at a point that appears to be -2 , then decreases and reaches a minimum at $x = 0$, before increasing again. Let us then try to determine ‘visually’ the intervals in which *f* is either concave or convex. The function *f* is first concave, up to a value x_0 somewhere between -2 and 0 ; then it becomes convex after x_0 . How can we determine x_0 ? To do this, we need to look not at the values of the derivative, but at how the derivative varies – that is, how the slope of the tangent to the graph of f' changes. Instead of examining the graphs of *f* and f' , it is technically more concise to simply examine the graph of f'' (see Figure 6). Here, $f''(x) = x + 1$. The second derivative f'' is therefore represented by a straight line with slope 1. It is strictly negative when $x < -1$, zero for $x = x_0 = -1$ and strictly positive when $x > -1$. The reader

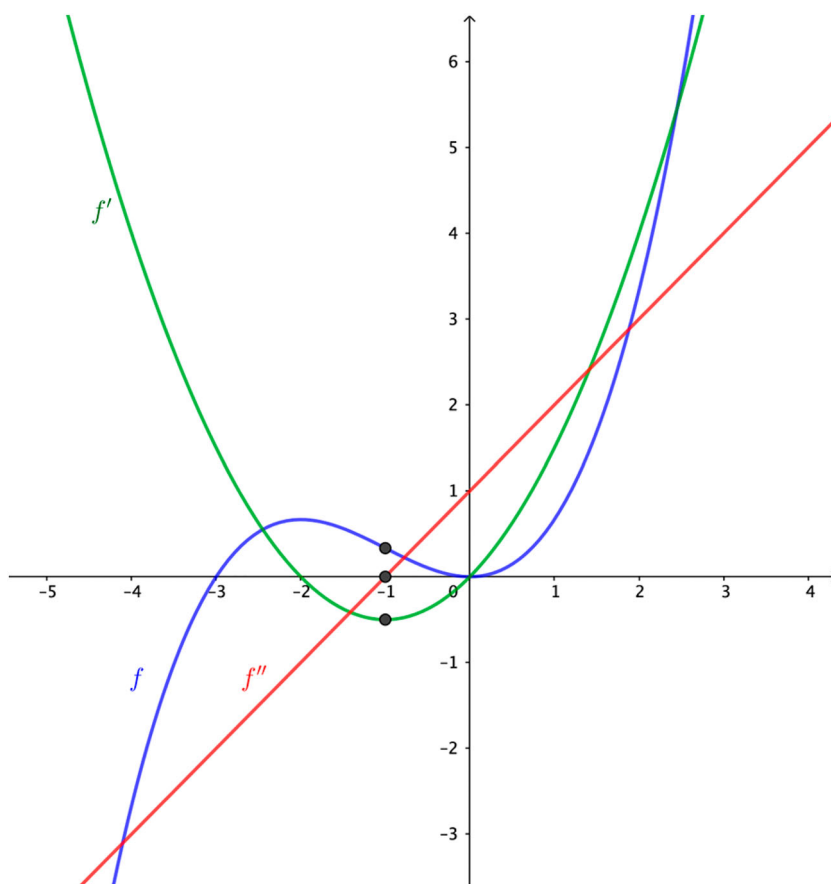


Figure 6. Graphs of f , f' , and f'' (adapted from Kalvø et al., 2021, p. 211). Reprinted with permission.

can examine two animated GIFs by clicking on the links below. Note that the first animation highlights the *values* (positive, negative) of f' while the second animation highlights the fact that f' is *increasing* or *decreasing*.

https://commons.wikimedia.org/w/index.php?title=File:Tangent_function_animation.gif&oldid=507127692

https://upload.wikimedia.org/wikipedia/commons/7/78/Animated_illustration_of_in_flection_point.gif

3.3.2. A praxeology of concave/convex functions in the school textbook

If we look at the treatment of concavity in *Patterns* as a certain *praxeology*, we can analyse it as explained in the following paragraphs.

The *type of tasks* T studied is formulated more allusively than explicitly. A task t of type T consists in determining the ‘curvature’ of the graph of a given function f and finding its possible turning point(s). This involves determining intervals of $\mathbb{R}$ on which f is either ‘concave down’ or ‘concave up’ by examining a ‘sign line’ as shown in Figure 5.

The notion of concavity is portrayed rather than properly defined. This is made possible, among other things, by a skilful linguistic ‘manoeuvre’, which is one of the keys to

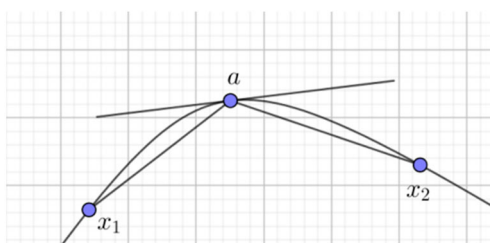


Figure 7. Chords and tangent to the graph of a function.

the didactic transposition carried out by the authors: as we have already pointed out, the words *concavity* and *inflection point* do not appear ('concave/convex' is replaced by the expression 'hollow side down/up', associated with the expression 'curvature'; 'inflection point' is replaced by 'turning point'). While the words 'concavity' and 'inflection point' are relatively opaque words in ordinary language, and therefore require comments, if not a precise definition, the expressions by which they are replaced belong to everyday language and are known to all, which authorises the authors not to say more about them.

The *technique* τ to perform a task $t \in T$ consists in calculating the second derivative f'' and studying its sign. Differentiating a function is a type of tasks T_1 and determining the sign of a function is a type of tasks T_2 . In essence, both T_1 and T_2 are assumed to have been studied beforehand and to be now largely routinised. The only new feature is that, given the function f , the derivatives f' and f'' must be calculated successively.

The *technology* θ of the technique τ is reduced to a minimum. One would expect for example that when f is concave down, the authors would point out to their readers that the slope of the tangent decreases. Instead, they invite them to observe, on the graph of f' , that f' is decreasing. Even more so, they do not care to mention the equivalence of various properties such as

- the slope of the tangent decreases
- the curve is below its tangents
- for any point a on an interval I on which f is defined, the function $r_a: x \mapsto r_a(x) = \frac{f(x)-f(a)}{x-a}$ decreases (an example where $r_a(x_1) > f'(a) > r_a(x_2)$ is given in Figure 7).

The third of these properties corresponds to Lemma 6.4.6 in *Calculus*, explained in Section 3.2.

Finally, about the praxeology of concave/convex functions in *Patterns*, we have uncovered that no (mathematical) *theory* Θ justifies the technology θ . This can be explained by two factors: first, θ is almost non-existent; second, the authors make no real attempt to justify what little exists of θ . We would like to state here what the theory, according to the ATD, would be, to wit, a system of statements (definitions, axioms, lemmas, theorems, corollaries ...) from which we can derive an acceptable justification θ . Let us suppose, for example, that we want to justify the fact that, when the derivative f' decreases, 'the curve is below its tangents'. To do so, consider a continuous function f on an interval I , and let $a \in I$ and $\varepsilon > 0$. As shown in Figure 8, we have: $f(a) + \varepsilon f'(a) - f(a + \varepsilon) = \varepsilon f'(a) - [f(a + \varepsilon) - f(a)]$.

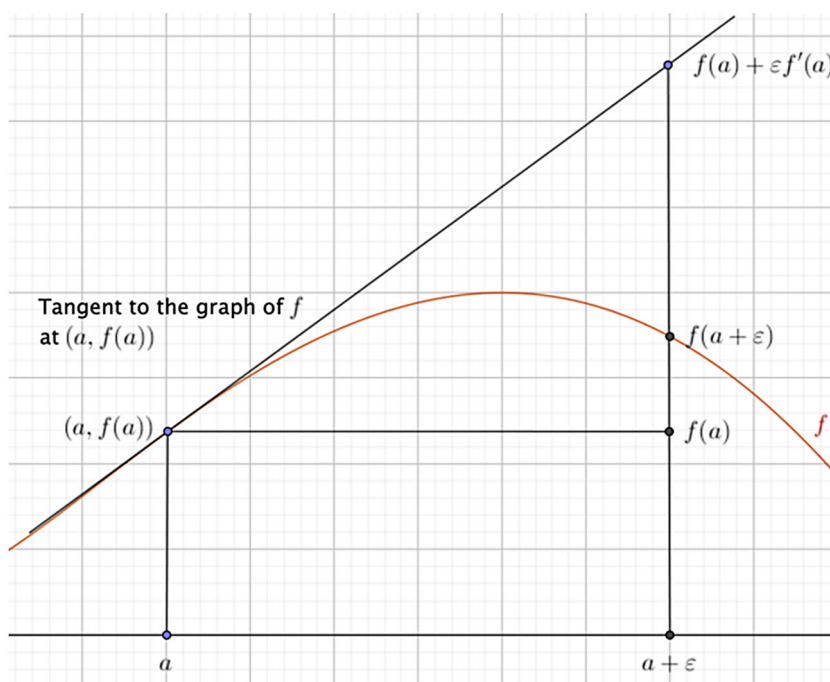


Figure 8. Decreasing derivative.

According to the mean value theorem, there exists $\gamma \in (0, 1)$ such that $f(a + \varepsilon) - f(a) = \varepsilon f'(a + \gamma\varepsilon)$. We thus have: $f(a) + \varepsilon f'(a) - f(a + \varepsilon) = \varepsilon f'(a) - \varepsilon f'(a + \gamma\varepsilon) = \varepsilon [f'(a) - f'(a + \gamma\varepsilon)]$. Since f' decreases, $f'(a) > f'(a + \gamma\varepsilon)$ and therefore $f(a) + \varepsilon f'(a) - f(a + \varepsilon) = \varepsilon [f'(a) - f'(a + \gamma\varepsilon)] > 0$. ■ In the case of this small investigation, we could look for the mathematical ‘principles’ that theoretically justify the mean value theorem and the tools used to establish it (e.g. Rolle’s theorem⁵).

3.4. Comparison of the praxeologies relating to $\mathcal{K}$ in the two textbooks

The analyses presented in the preceding subsections demonstrate that the secondary school textbook features minimal mathematical *logos*. This contrasts with the presentation of the same theme in the university textbook, where we found a *logos* block consisting of definitions and proved results. In the school textbook, the notion of concavity has been substituted by an *application* of the notion of second derivative and, consequently, there is an exclusion of questions where concavity could have come into play. The two other mathematics textbook series mentioned above (Borgan et al., 2021; Oldervoll et al., 2021) have a very similar treatment of concavity, using exactly the same notions as *Patterns*.

The main fact is that, whereas in the university presentation, the *graphical* notion of concavity is *mathematised*, in the secondary school textbook it remains non-mathematised (or weakly mathematised): concavity is *to be seen* on the graph of the function. At best, this *visual* property is simply translated by saying that the slope of the tangent to the curve

decreases or increases, which in turn is translated mathematically by the sign of the second derivative. The crucial gain is obvious: the subtle work required to mathematise the graphical notion of concavity is avoided, so that its presentation is more easily accessible to a wider audience.

The analytic results presented in this section will be discussed in Section 5. To facilitate this discussion, we need a reference model from which the organisations of knowledge we consider in this paper can be derived by transposition. In the subsequent section, we endeavour to construct a model of this nature.

4. A tentative ‘archeorganisation’ for concave/convex functions

We now turn to an examination of our second research question, *RQ2*, introduced in Section 1. The inquiry proceeds by sketching out as much as possible, within the restricted scope of this paper, what we call an *archeorganisation* for concave/convex functions. This reference model could be considered authoritative and instrumental in mathematics teacher education institutions by satisfying the needs in terms of utility and legitimacy of teacher educators and prospective teachers. The notion of an archeschool, explained in Section 2.3, is central to understanding the concept of an archeorganisation as it forms the foundational reference from which the particulars of praxeological utility and epistemic legitimacy are derived. In essence, the archeschool is both a ‘meta-institution’ and a conceptual construct that provides a unifying standard for the creation, evaluation and dissemination of knowledge within the noosphere of various educational and organisational institutions.

Let us start. While some mathematical objects (e.g. the notion of irrational number) seem to belong only to mathematics, the notion of concavity/convexity – hereafter generally represented by *concavity* – is present in many fields of institutional activity, including everyday life, architecture, optics, economics, anatomy, among others. It is advisable that the targeted education highlights both the praxeological similarities and the praxeological differences that can be observed in the institutions using these concepts. The archeorganisation to be developed must therefore allow for the emergence of the ‘mathematical choice’ within a variety of observable institutional choices.

Before we present a (mathematical) theory of concave/convex functions, we provide concise accounts of how the notion of concavity is used in different institutions. This theory builds on Chevallard’s (in press) model based on three generic questions about any object $\mathcal{K}$: (1) What is $\mathcal{K}$ made of? (its *structure*); (2) how does $\mathcal{K}$ work? (its *functioning*); and (3) what is $\mathcal{K}$ used for? (its *utility*).

4.1. ‘Concave’ and ‘convex’ in various institutions

According to Merriam-Webster ([n.d.a](#), Definition 1), the term *concave* refers to a shape that is ‘hollowed or rounded inward like the inside of a bowl’. On the other hand, the term *convex* describes a shape that is ‘curved or rounded outward like the exterior of a sphere or circle’ (Merriam-Webster, [n.d.b](#), Definition 1a). Thus, the words ‘concave’ and ‘convex’ are direct antonyms, which in everyday language mean that a concave shape is hollow and can be filled, while a convex shape creates a dome and is bulging.



Figure 9. Concave roof (Santos, 2016). Courtesy of BMDesign Studios. Reprinted with permission.

These connotations are used in various domains, for example in architecture. ‘Concave Roof’ is a roof system with steeply sloping sides reminiscent of a bowl (see Figure 9). Its innovative design has been created by the Iran-based BMDesign Studios, intended to collect rainwater in arid regions, where water scarcity could have severe consequences (Santos, 2016).

By contrast, the Dome of the Rock, a well-known landmark in ancient Jerusalem, features a convex roof, which is a distinctive characteristic of its architectural design, shown in Figure 10.

Having considered these examples from architecture, let us make a quick comment on optics. A *concave lens* is such that when parallel rays of light pass through it, the refracted rays diverge so they appear to come from one point called the principal focus (Convex & Concave Lenses, Concave Lens section, 2020). Conversely, a *convex lens* is such that when parallel rays of light pass through it, the refracted rays converge at one point called the principal focus (Convex & Concave Lenses, Convex Lens section, 2020). Figures 11 and 12 display the two types of lenses, where we see that the characterisation of their shapes is consistent with the everyday connotation presented above.

A remark is warranted here since some readers may find what is presented so far to be inconsistent with their understanding of concave/convex functions based on their mathematical knowledge. The concept of concavity/convexity, used in various fields to refer to shapes that are either hollow or domed, is dependent upon the observer’s viewpoint. Specifically, it varies based on the observer’s position relative to the object, whether it be in a vertical direction, seen from above/below, or in a horizontal direction, seen from the left/right. For example, when viewed from above, the Concave Roof design is concave (hollow), and the roof of the Dome of the Rock is convex (domed). In today’s mathematical terminology, these shapes would be characterised as convex and concave, respectively (i.e. the opposite), since the common viewpoint of the ‘observer’ in mathematics is from below.



Figure 10. Convex roof ('Dome of the Rock', 2019). Courtesy of Andrew Shiva.

(We will come back to this in Section 4.2.) Furthermore, the lens in Figure 11 is concave (hollow) and the lens in Figure 12 is convex (domed), regardless of being viewed from the left or from the right, which is due to their vertical symmetry about a central axis. In general, as much as the observer's position in relation to objects is important for determination of concavity, the position of the objects themselves is critical. While some objects have a static positioning (e.g. the architectural objects in Figures 9 and 10), other objects, or graphic representations of them, have a dynamic positioning (e.g. the lenses in Figures 11 and 12). It can also be noted that for objects having symmetrical shapes, the characterisation of concavity does not change when the observer's viewpoint in the observational direction changes.

The issue of the observer's viewpoint was considered as early as 1941 in a publication by the economist Adolf Kozlik (1912–1964), where he wrote:

The terms convex and concave are used by economists in their descriptions of curves in different, sometimes confusing ways. Although nomenclature is a matter of convention and taste (if only definitions are properly stated and adhered to) a uniform language has great advantages. It may, therefore, be recommended to standardize the terminology concerning the shape of curves. (Kozlik, 1941, p. 103)

Kozlik then discusses three ways of addressing the matter:

These three expressions are commonly found. Curves are (a) concave or convex 'to the origin'; (b) concave or convex 'upward' or 'downward'; (c) concave or convex 'from above' or 'from below'. (Kozlik, 1941, p. 103)

We will leave aside here the expression (a) as used by economists. On the other hand, here is Kozlik's comment about the expression (b):

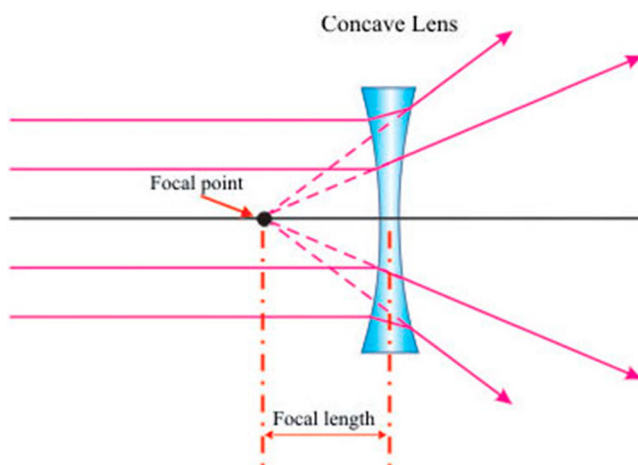


Figure 11. Concave lens (Convex & Concave Lenses, Concave Lens section, 2020).

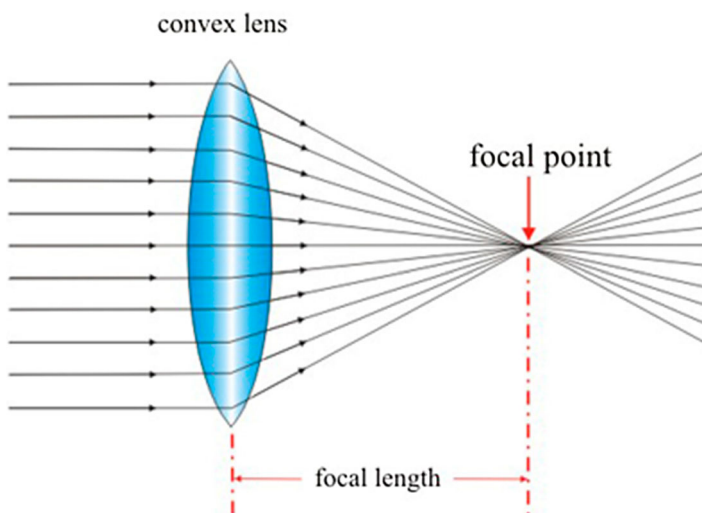


Figure 12. Convex lens (Convex & Concave Lenses, Convex Lens section, 2020).

Terminology (b) is ambiguous because it is not clear without further specification whether the terminology means that the convex or concave side of the *curve looks upward or downward* (Viner, Waugh) or that the curve appears to be convex or concave as the *observer looks upward or downward*. (Kozlik, 1941, p. 104)

Kozlik then arrives at expression (c): ‘Terminology (c) is the one used by all mathematicians and therefore mathematical economists and most of the other economists’ (p. 104). For various reasons, he advocates the universal adoption of this terminology. In the reference model of a theory of concave functions, we will therefore retain expression (c). However, we will differ from Kozlik on one point. He recommends that the expression ‘convex or concave from *above*’ should be adopted because it is more commonly used than ‘concave

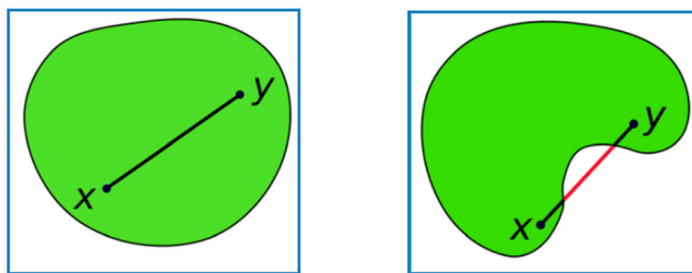


Figure 13. Convex set (left) and non-convex set (right) ('Convex Set', 2023).

or convex from *below*' (p. 104). This recommendation is not in line with the usual mathematical terminology of today, where the observer's viewpoint is from below, as commented above.

4.2. A mathematically refined theory of concave/convex functions

An archeorganisation cannot be decreed. It is the result of a convergence process in a system of related institutions, such as teacher education institutions. So, what kind of archeorganisation could be aimed at here? The developments in Section 4.1 lead to a first element of an answer: it is appropriate to show that the mathematical relation to the notion of concave/convex functions is linked with, but differs from, the notion of concave/convex as used outside mathematics.⁶ Let us now turn to structure, functioning and utility.

4.2.1. Structure of the theory of concave/convex functions

In this subsection, we will synthesise some mathematical accounts of the notion of concave/convex into a concise summary.

First of all, a notion that is central, even if it is often put aside by some authors, is the notion of *convex set*: a subset C of $\mathbb{R}^n$ (where n is an integer ≥ 1) is convex if, for any pair (x, y) of elements of C , the line segment $\overline{xy}$ is contained in C , as shown in the left illustration in Figure 13.

This being the case, we can define the notion of a *concave* function on an interval $[a, b]$ by saying that the set of points *on or below* its graph, that is, $\{(x, y) \mid a \leq x \leq b, y \leq f(x)\}$, is a *convex* set. Similarly, for a *convex* function on $[a, b]$, the set of points *on or above* its graph, that is, $\{(x, y) \mid a \leq x \leq b, y \geq f(x)\}$, is a *convex* set. In Figure 14, the graph on the left is therefore that of a concave function, the graph in the middle is that of a convex function, while the graph on the right is that of a function which is first concave and then, from a certain point, convex.

It can be proved (see Choquet, 1964/1966, pp. 159–161) that a concave or convex function f on an *open* interval I is continuous at any point of I and has at each point a left derivative f'_l and a right derivative f'_r , the set of points where $f'_l \neq f'_r$ being at most countable; moreover, when f is concave, these derivatives are decreasing; they are increasing when f is convex. We will limit ourselves to quoting the following result:

Let f be a finite numerical function on an open interval I , and let D be an at most countable subset of I . In order that f be convex, it is necessary and sufficient that f be continuous,

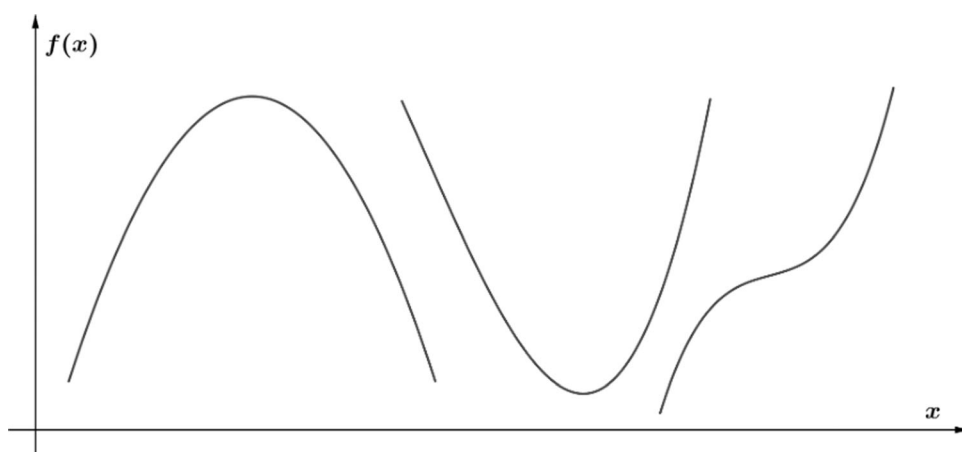


Figure 14. Bending of graphs: concave (left); convex (middle); first concave and then, from a certain point, convex (right).

have a right derivative f_r' at every point of $I \setminus D$, and that f_r' be increasing on $I \setminus D$. (Choquet, 1964/1966, p. 161)

After presenting this proposition, the author continues with the next, well-known corollary:

If a function f has a second derivative f'' at every point of the open interval I , the convexity of f is equivalent to the relation $f'' \geq 0$. (p. 162)

Of course, if f has a second derivative f'' , f is concave if and only if $f'' \leq 0$. Another definition of the pair convex/concave is given by Izrail S. Gradshteyn and Iosif M. Ryshik (1943/2015, p. 1074):

A function $f(x)$ is said to be **convex** on an interval $[a, b]$ if for any two points x_1, x_2 in $[a, b]$,

$$f\left(\frac{x_1+x_2}{2}\right) \leq \frac{f(x_1)+f(x_2)}{2}.$$

A function $f(x)$ is said to be **concave** on an interval $[a, b]$ if... the function $-f(x)$ is convex [on] that interval.

This definition states that a function $f(x)$ is convex on an interval $[a, b]$ if the average of the function values at any two points x_1 and x_2 on that interval is greater than or equal to the function value at the midpoint of x_1 and x_2 . It results from this that the graph of the function lies below the chord connecting any two points on the interval (Choquet, 1964/1966, pp. 162–163). If the function satisfies the opposite inequality, where the graph of the function lies above the chord, then the function is concave on the interval.

When we compare Lindstrøm's (2016) definition of convex/concave functions, given in Section 3.2, to the one given here by Gradshteyn and Ryshik (1943/2015), we see that they are similar. However, while the former is a geometric definition, which provides a visual way of understanding the notions of convex and concave functions, as it explicitly refers to the position of points on the graph of the function, the latter definition uses algebraic criteria to define the same notions, based on comparison of the function's midpoint value with the average of its values at two points on the interval in question. Some authors of calculus textbooks, such as Robert A. Adams and Christopher Essex (2018) and Claudia Neuhauser

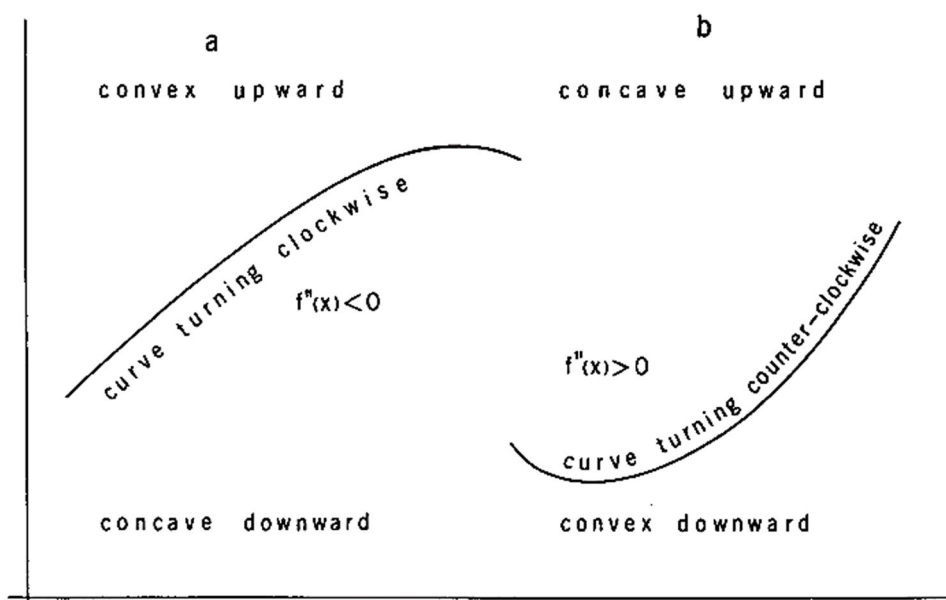


Figure 15. Concave/convex as illustrated by Batschelet (1979, p. 267). Reprinted with permission.

(2011), use the pair concave up/concave down instead of the pair convex/concave. In contrast, Edward Batschelet (1979) provides dual descriptions for each term, a categorisation based on which direction the curve turns (see Figure 15). Concave downward is equivalent to convex upward (corresponding to a clockwise turn), and convex downward is equivalent to concave upward (corresponding to a counter-clockwise turn), denoting, respectively, concave and convex functions in mathematical terminology, as explained in Section 4.1. We note that Batschelet's 'downward' consideration of curves aligns with the perspective of viewing curves from below, clarified as the observer's viewpoint in mathematics.

We will go no further here as to the *structure* of the object under consideration.

4.2.2. Functioning of the theory of concave/convex functions

The functioning of the theory of concave/convex functions depends on what the theory is used for: functioning and utility are related. But we can look at some relatively generic ways of functioning, exemplified through some 'rules' presented, for example, by Batschelet (1979). The first one is about concave functions:

For practical applications the following rule is useful: A function $y = f(x)$ reaches a maximum value at x_0 if $f'(x_0) = 0$ and $f''(x_0) < 0$. (p. 267)

A little further on, he states the corresponding 'rule' for convex functions:

A function $y = f(x)$ reaches a minimum value at x_0 if $f'(x_0) = 0$ and $f''(x_0) > 0$. (p. 267)

Then he goes a step further by showing that, in some cases, it is possible *not to study* the sign of f'' , using the following 'rule': if x_0 is inside the interval under consideration, f reaches its minimum on this interval at x_0 if $f'(x_0) = 0$, $f'(x) < 0$ when $x < x_0$ and $f'(x) > 0$ when $x > x_0$. Batschelet implements the latter 'rule' on finding the minimum value of

the function $f(x) = \frac{2x^2-1}{x^2+1}$, by showing that $f'(x) < 0$ for $x < 0$ and $f'(x) > 0$ for $x > 0$, which implies that $f(x)$ reaches a minimum at $x_0 = 0$, where $f(x_0) = -1$ (Batschelet, 1979, p. 271).

It is worth noting that in Batschelet's book, the discussion of concavity of curves is located in the subchapter 'The Second Derivative'. This treatment is the same as in the textbook *Patterns* (Kalvø et al., 2021), as commented in Section 3.3.1. Such a placement indicates that while concavity is an important topic, it is primarily considered as an application of the second derivative rather than as a concept in its own right.

4.2.3. Utility of the theory of concave/convex functions

The question of the *utility* of a mathematical object $\mathcal{K}$ is rarely addressed in mathematical presentations relating to $\mathcal{K}$, which concentrate on the *structure* of $\mathcal{K}$ and, secondarily, on its *functioning*. In Subsection 4.1, we presented some practical examples of $\mathcal{K}$, denoting concave and convex functions, showcasing their utility in disciplines such as architecture and optics. Here, we will introduce two further examples, one from economics and the other from elementary real analysis, which highlight the mathematical significance of the utility of $\mathcal{K}$.

In the field of economics, two pivotal concepts are *strict concavity* and *utility function*. A function $f(x)$ is strictly concave over an interval if, for any two distinct points x_1 and x_2 within the interval and for any λ in the open interval $(0, 1)$, the following strict inequality holds: $f(\lambda x_1 + (1-\lambda)x_2) > \lambda f(x_1) + (1-\lambda)f(x_2)$. This means that the line segment connecting $(x_1, f(x_1))$ and $(x_2, f(x_2))$ lies strictly below the graph of the function except at the endpoints (i.e. there are no straight-line segments on the graph of f). While both concave and strictly concave functions ensure that there is a maximum point, strict concavity guarantees the uniqueness of the maximum, which is a valuable property when it comes to ensuring the stability of equilibria and the predictability of economic models.

Within risk aversion theory, a utility function is a model of how a consumer's satisfaction (utility) increases with wealth but at a decreasing rate. This phenomenon occurs because, given the principle of diminishing marginal utility, each successive unit of wealth yields a smaller increment in satisfaction than the previous one (see, e.g. Mehta, 1998, for a detailed discussion of the utility function concept). Strict concavity holds particular significance within the economic theory of risk aversion, as exemplified in the seminal work of Pratt (1964).

A strictly concave utility function implies that an individual prefers certain outcomes over uncertain ones with the same expected value. However, this relationship becomes more complex with state-dependent preferences, as demonstrated in the paper 'Concavity, stochastic utility, and risk aversion' by Jarrow and Li (2021). In their example (pp. 319–320), they define a utility function $U(x, \omega)$ that is strictly concave and increasing in wealth x , yet the individual's preferences may not exhibit risk aversion due to the utility's dependence on the state ω . The utility function changes at the expected state $\omega = 0.5$, with different utility values for states below and above this point. This state dependence can lead to situations where, contrary to the usual implications of concavity, an individual might prefer a risky outcome to a certain one, because the expected utility of the risky outcome is higher. The authors express that this example underscores the importance of considering how external

conditions or ‘states of the world’ can influence preferences and decision-making under risk.

We hope that this succinct description of strict concavity has illuminated its utility in economics, even for readers less familiar with the mathematics of this discipline. Now, let us consider another example, a straightforward use of $\mathcal{K}$ in the field of elementary real analysis. On the website ‘Bibliothèque des mathématiques’ (Library of Mathematics), the following exercise can be found:⁷ ‘Show that, for all $x \in [0, \pi/2]$, we have $\frac{2}{\pi}x \leq \sin x \leq x$ ’ (‘Solved Exercises – Convexity’, Exercise 6, n.d.). The proposed solution is as follows:

Since $(\sin)'' = -\sin$, the sine function is concave on $[0, \pi/2]$. Its representative curve is therefore, on this interval, below its tangent at 0, and above the chord joining $(0, \sin 0)$ to $(\pi/2, \sin(\pi/2))$. Now, the equation of this tangent is $y = x$, and the equation of this chord is $y = \frac{2}{\pi}x$. We obtain exactly the result we asked for.

Before stopping these considerations on the largely ignored question of utility (let us repeat: most presentations give the lion’s share to structure, as little as possible to functioning, and pay almost no attention to utility), we will only add a question that could be the subject of inquiries by student teachers: ‘Why does the notion of a concave function seem to occupy a significant place in machine learning?’ In other words, what is the utility (or utilities) of this notion in machine learning? Of course, a similar question about *optics* could also be the subject of such inquiries.

4.3. Some concluding remarks on the envisioned archeorganisation

Regarding the theory of concave and convex functions (the knowledge object $\mathcal{K}$ that we consider), our goal is to contribute to the institutional (and therefore collective) emergence of a praxeological organisation that would function as an ‘archeorganisation’ in institutions educating secondary mathematics teachers. This framework should thus be, at least for a few decades, the de facto authoritative and encompassing source of the organisations actually living in these institutions.

We have pointed at praxeological similarities and praxeological differences that can be observed in various fields using the notion of concavity. These differences have revealed a contradiction between the *mathematical* notion of concavity that we intend to teach students, and the *ordinary* relation manifested in social life, viewed as an all-encompassing institution.

The questions concerning structure, functioning and utility of $\mathcal{K}$ are interrelated. Given the nature of the system of institutions targeted here – mathematics teacher education institutions – we have addressed structural aspects at different levels of mathematical sophistication: While Batschelet’s (1979) theory is clearly mathematically constrained, the theory of Choquet (1964/1966) is mathematically more refined. There are, of course, other possible distinctions than those we have made here on the structure of $\mathcal{O}$. Due to limitations of space, the questions of functioning and utility have been only minimally addressed in the preceding subsections. Even so, it is essential to pay attention to the possible praxeological discrepancies that transposition processes not regulated by an adequate archeorganisation can lead to. Such a situation of institutional fragmentation of knowledge is likely to constitute an obstacle to mathematical, and more generally praxeological, exchange between, in particular, university education and secondary education.

5. Discussion and conclusion

5.1. Diachrony and synchrony in didactic transposition analysis

Let us consider a praxeological element $\mathcal{K}$ present in an institution I . We defined in Section 2.2 the genealogy of $\mathcal{K}$, that is, the n -tuple $(\bar{I}_1, \bar{I}_2, \dots, \bar{I}_n)$. The element $\mathcal{K}$ can be subjected to a praxeological analysis with a view to determining, when possible, which components of $\mathcal{K}$ are derived from which of its ancestors $\bar{I}_m$ ($1 \leq m \leq n$) and which components present in such and such of its ancestors *no longer appear* in $\mathcal{K}$, and why. Such a praxeological analysis is called a *diachronic* praxeological analysis (or a praxeological analysis *in diachrony*). Elsewhere, we carried out such a diachronic study as concerns the historical vanishing of *parameters* from elementary algebra, highlighting what, in the genealogy of elementary algebra, caused this ‘catastrophe’ (Strømskag & Chevallard, 2022). Although such events are not rare and may have various causes, it should be remembered that the type of event that is probably most frequent in a praxeological genealogy consists of *updates* brought about by changes in the praxeological environment of $\mathcal{K}$ ’s ancestors. (An example of such updates was given at the outset of this paper, not in relation to a mathematical object, but to an object of the ATD, the notion of ‘body of knowledge’.)

In all cases, a diachronic praxeological analysis presupposes one or more *comparative* praxeological analyses for contrasting pairs $(I, \mathcal{K})$ and $(I', \mathcal{K}')$. Comparative praxeological analyses are an essential type of tasks in the analysis of didactic transposition processes. But this type of tasks is also at the heart of *synchronic* praxeological analysis, which requires us to compare pairs $(I, \mathcal{K})$ and $(I', \mathcal{K}')$ even when neither of them is a proven ancestor of the other. This is what we have tried to do in Section 3, by contrasting two textbooks in their treatment of the notion of concavity/convexity.

5.2. The fragmentation of knowledge: an exemplary case

Unlike certain objects, which live specifically in a given type of institutions, the notion of convexity/concavity lives in a plurality of institutions that cannot be reduced to a single type. In concrete terms, which institutions should a person wishing to learn more about convexity/concavity turn to? Architecture? Optics? Botany? Economics? What we have here is a tenacious institutional *fragmentation* of knowledge. What *apparently* limits the effects of this heterogeneous plurality might be called ‘institutional solipsism’, which leads an institution to generally ignore most of other institutions. The institutions involved each have their own truths, their own ways of saying and doing things – in short, their own praxeologies – about the subject under consideration. For example, most readers have probably never come across the word *concaver*. But who uses it, and where?⁸

As we have seen in Section 3, with the examples of the *Calculus* and *Patterns* textbooks, even in institutions which are close to each other and which are supposed to be interconnected, fragmentation can be strong, as if no dialogue between institutions can exist. Whereas the university textbook includes a substantial *logos* block relating to concavity, complete with definitions and proven results, the secondary school textbook includes as little mathematical *logos* as possible. The benefit is evident: by circumventing the process of mathematising the graphical notion of concavity, its presentation is made more widely available to a broader spectrum of students.

Another gain stems from an ‘iron law’ of curriculum crafting: a new item benefits from appearing as an ‘application’ of an established item – here the notion of second derivative. Is there a loss? Yes, there is. Whereas at university, under appropriate regularity conditions, one can *prove* that, if a function is concave down, its second derivative is negative, and *conversely*, at the secondary level, for lack of a *mathematical* definition of concavity, students will miss this particular opportunity to get a simple, founding experience in their mathematics education: tackling a theorem, and then its reciprocal. In the transition from secondary school to university, there happens a breach of epistemic contract and a consequent didactic disruption. Didactic transposition thus surreptitiously makes its mark, and sometimes takes its toll, on students’ and teachers’ *praxis* and *logos* by distorting and, often, damaging the mathematical equipment which could be available to them.

In addition to this deficit in mathematics education, which could be *partly* remedied by the systematic practice of stating explicitly that such and such statement is accepted without proof, rather than implying that it is self-evident, the phenomenon of institutional fragmentation of knowledge and its consequences must be emphasised. It is not wrong in itself to try to partially demathematise a praxeological organisation. This is what Batschelet (1979) does, for example, by showing that, in some cases, one can do without the calculation of the second derivative f'' . But in the case under consideration, that of secondary schools, it seems that a ‘praxeological isolate’ will form, with its own life, and whose links with other institutions will be severed. This will inevitably lead to some form or other of curricular obsolescence. In such a case, the privileged place for a useful curricular ‘defragmentation’ and ‘rejuvenation’ seems to be none other than teacher education institutions, provided they refer collectively to an appropriate archeorganisation. And it is to such an epistemological and cultural (re)construction that we have endeavoured to contribute here.

5.3. Towards an archeorganisation of knowledge?

To conclude, we will briefly summarise the constituents of curriculum design which may lead to an archeorganisation as we have portrayed it. We will do this in the form of a moral fable featuring a secondary school teacher $\hat{y}$ who, considering what she may have learned (and *not* learned) when she was a student in upper secondary school or a student teacher at university about the notion of convexity/concavity, imagines suing these institutions for ‘epistemic ill-treatment’, a concept of her own invention. She therefore decides to draw up a three-point ‘indictment’ concerning, respectively, the structure, the functioning and the utility of the praxeological complex that she has had or has not had the opportunity to experience during her years of study. This (imaginary) indictment consists of three straightforward questions, illustrated by a very few examples.

On the subject of *structure*, she is stunned to have never personally come across a statement like the one she found recently in Choquet’s old book on topology (1964/1966, p. 161): if $f: I \rightarrow \mathbb{R}$, where I is an open interval, then f is convex if and only if (1) f is continuous, (2) f has a right derivative f_r' except on an at most countable subset D of I and (3) f_r' is increasing on $I \setminus D$. As for the *functioning* of the notion of convexity/concavity, she is also surprised that she has only heard of the second derivative, and not, for example, of this easy ‘rule’ (Batschelet, 1979, p. 271): if $x_0 \in I$, f reaches a minimum at x_0 if $f'(x_0) = 0$, $f'(x) < 0$ when $x < x_0$ and $f'(x) > 0$ when $x > x_0$. In this respect, she has also detected several ‘functional’ properties, such as the fact that, if $f_1, \dots, f_n$ are convex,

then $\max\{f_1, \dots, f_n\}$ is convex; or that, if f is convex, then $e^{f(x)}$ is convex, etc. ('Convex Function', 2023).

This leads her to raise the general question of *utility*. When, for example, does one need to use the fact that $e^{f(x)}$ is convex? More widely, what can be the utility of the mathematical theory of convexity and concavity in economics, physics, architecture, etc.? Can mathematics be at the heart of a defragmentation of knowledge about this subject? She realises that she cannot answer these questions without further appropriate inquiries. Of course, this undertaking exceeds her personal possibilities, although she believes she could valuably contribute to it. Her fantasy is only an outline for what should be a precious democratic debate, as should be the case with all curricular issues, which can only be developed within the relevant noosphere (which, in principle, excludes no one), striving to build up a sustainable archeorganisation. It should be obvious that the question of 'epistemic ill-treatment', the study of which has been illustrated here in the particular case of concave and convex functions, arises for *all* objects whose study at secondary school or university is contemplated. Hence, our study paves the way for a systematic examination of epistemic fidelity, openly inviting future research to explore these dimensions for other mathematical topics.

Notes

1. Quotations from these textbooks have been translated into English by the first author.
2. Figures taken from *Calculus* used in this paper are renumbered to have continuous numbering, and they are given titles by the authors (figures are untitled in the source).
3. See the *Mean Value Theorem* at MathWorld: <https://mathworld.wolfram.com/Mean-ValueTheorem.html>
4. In Norwegian, the use of 'bøypunkt' would be preferable to 'vendepunkt' when seeking a colloquial equivalent for the concept of inflection point. 'Bøypunkt' translates to 'bending point' in English, and more accurately captures the nature of the point on a curve where concavity changes.
5. See *Rolle's Theorem* at MathWorld: <https://mathworld.wolfram.com/RollesTheorem.html>
6. This point is further illustrated on the blog *Prawfs Blawg*: <https://prawfsblawg.blogs.com/prawfsblawg/2009/12/convexed.html>
7. The French text has been translated into English by the second author.
8. For an answer, see for example at <https://en.wiktionary.org/wiki/concaver>.

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